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Technical note

Acoustic comfort evaluation for a conference room: A case study

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ABSTRACT

This paper presents the evaluation of acoustic comfort for a conference room located in a sample of newly built public buildings. The inside and outside ambient noise, interior sound insulation and the reverberation time are measured according to international standards. Regarding acoustic requirements for spaces intended for speech communication, the results obtained from measurements are compared to those given as guidelines and reference values, and which are recommended by some national and international standards. This comparison reveals the existence of a poor acoustical quality in the conference room, and which is caused by a relatively excessive level of the ambient noise, low insulation between the technical and conference rooms and high value of the reverberation time. This inconvenient situation is attributed to deficiencies in concept details at the early building stages, and which is principally related to the inappropriate consideration of the acoustical aspects that the building is expected to fulfill. Some recommendations are consequently given and discussed for remedying this situation and improving the acoustics of the building.

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1. Introduction

Good acoustics is essential for comfort and productivity in workspaces. In closed spaces, where oral information exchange or learning processes involve intensive verbal communication, a good design is required for optimizing the primary function the spaces are intended for.

However, from everyday experience, each of us has surely noticed that classrooms, conference rooms, lecture theaters, or halls of worship, can be acoustically satisfactory, as well as unsatisfactory. This is translated, in general, by judgments such as "this room has better acoustics than the other one," or "we come to follow lectures in this room more comfortably than in the other room"...etc. Similarly, we give sometimes different acoustical assessments for the same room. These subjective impressions depend on the position of the person in the local and also may vary from one person to another. Indeed, two people can have different impressions concerning the same sound signal at the same place in a particular room. For this kind of subjective judgments, scientists are working on elaborating some objective or measurable parameters in order to be able to guide the architect in his efforts of conceiving an adequate design to a building fulfilling some specific function [1–5]. These parameters are necessary for a more comprehensive evaluation of the acoustic quality of rooms.

For spaces dedicated to oral communication the factors that can influence highly the speech intelligibility are the reverberation and the signal-to-noise ratio (S/N) of the speech in comparison to the ambient noise [1,6–11]. A noise source can be either external such as traffic noise, or internal including even the ventilation and air conditioning system (HVAC), the occupants themselves, the various machinery in the room and the eventual vibrating surfaces at the boundaries of or within the room [10,12,13]. An optimized reverberation reinforces early reflected energy of sound, which arrives at a receiver position at times less than 50 ms from that of the direct sound, and increase speech intelligibility [14,15]. On the other hand, the late energy, which arrives at a receiver position at times later than 50 ms, may mask the direct sound, hence effectively increasing the background noise, and decreasing speech intelligibility.

The results of several studies conducted around the world focusing on the subject of acoustics comfort, have concluded that the main reason for acoustic discontent is related principally to the lack of perception of the problem in the early phases of conception [9,16–20]. To avoid these drawbacks, different recommendations and guidelines have been established by different national and international organizations. These are used for helping during the phases of intervention under the construction or refurbishment

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project with the aim to provide reference values for different parameters that may influence acoustical comfort [21–26].

The goal of the present work was to evaluate the acoustical conditions, by measuring of ambient noise level, interior wall insulation and reverberation time, on a sample of learning spaces within a sample of lightweight design building.

The building is designed to be the new polyvalent, administrative and training, pole center of a national company. It incorporates different kind of work spaces such as offices, meeting and conference rooms. It shall host the survey and control system of the center region electricity network.

In this constructions, the frame is designed of metal according to requirements of national standard modules [27–29]. To achieve requirements relative to the energy efficiency and internal comfort, the separate walls are made of double-plasterboard spacing with glass wool. Furthermore, a several central systems for ventilation and air conditioning are installed. The external facades are on glass in order to favor an ambient lighting and also for esthetic considerations.

This study is a part of a procedure for assessing comfort in the building, identifying and proposing corrections for possible design defects. It can be of great importance given that the evaluated design is duplicated and intended to be duplicated in many other projects in other departments. The measuring methods and the measurement chain will be exposed in the first chapter. A brief description of the space under study is the objective of second chapter. The results obtained in different conditions will be presented, analyzed and discussed in the third chapter.

2. Materials and methods

The acoustic system of Brüel and Kjaer, B&K, building was used for taking measurements. It is a complete measurement system that covers the five diagnostic categories of sound characteristics through measuring the sound pressure according to ISO 717-1 [30], ISO 717-2 [31]. This system consists of the following elements:

- The Investigator 2260 system associated with BZ 7204 B&K and BZ7210 software.
- A B&K power amplifier, Ref. 2716.
- A B&K omnidirectional sound source, OmniPower, Ref. 4296.

2.1. Measurement of ambient noise

In practice, measurement of Sound Pressure Level (SPL) is the common used method to evaluate acoustic comfort with regard to ambiance noise, and the A-weighted sound level, L_A, is generally used in international standards for surveying if a measured noise level satisfies a specific requirement. However, L_A lacks specific spectral information and it can be misleading in evaluating noise. Hence, two different spectra can give the same numerical value, but be of quite different subjective characteristics. Thus, a more detailed procedure can be taken. One of the most useful methods to evaluate the acoustic comfort with taking into account the proposed activity within the local is to compare the measured SPL to that of a noise with known shape of sound spectrum. In fact, Noise Rating Curves, NC curves, are used as a means for providing a recommended noise criterion in rooms for various uses [7,13,32–35].

Acoustic measurements were planned as follows:

- External ambient noise
- Internal ambient noise, HVACs operation "off".
- The average level of noise in the conference room under normal operation conditions of HVACs.
- The level of HVACs noise in the equipment room in normal operation conditions.

The B&K 2260 Investigator have been used. The measurements were done during day time. The integration time was taken as three minutes for each measurement point. The results have been collected using Noise Explorer 7815: a Windows®-based software package for downloading, viewing and reporting noise data measured using the Brüel & Kjær hand-held instruments.

2.2. Reverberation time measurement

The reverberation time measurements were taken at three different positions in the room, and according to the standard requirements [36], the reverberation time can be correctly measured if three readings are taken at each of the measurement positions.

The data is transferred to a PC using the software Qualifier 7830. The quadratic pressure decay curves are plotted and the mean of reverberation times was calculated for each frequency band.

3. Description of the evaluated design

The structure of the building is made of a metal frame (see Fig. 2), containing mainly offices and rooms used for arranging conferences and meetings or for organizing continuous training. The facades are made of metal-glass combinations. Outside view of the site is shown in Fig. 1.

For air conditioning in the building several HVAC systems are installed. On the first floor, double HVAC (CTA in Fig. 1) are placed close to a conference room (see Fig. 1). The door of the room containing the HVAC systems, and which is of an ordinary type, leads directly to the conference room. The partition walls consist of two plasterboard BA13 separated by a 10 cm spacing filled with glass wool. The plan view and dimensions of the room are shown in Fig. 2.

The lateral surface walls of the conference room where the measurements have been conducted, are made of glass (38 m^2), plasterboard (106 m^2), and plywood (5.5 m^2) for the doors. Plasterboard Suspended Ceilings integrated in them lighting and air vent, and the marble slabs flooring totaled a surface area of around 97.5 m^2. 70 seats with low upholstering were also installed there.

Fig. 1. Outside view of site.

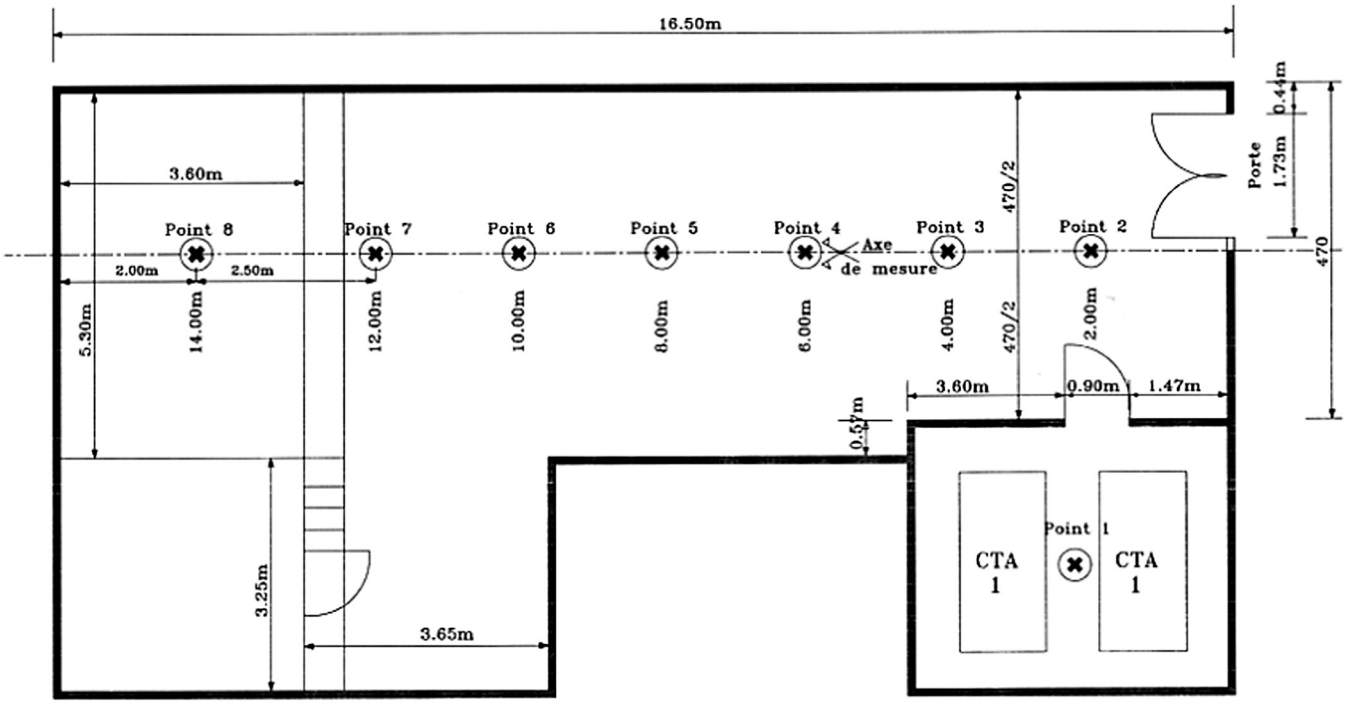

Fig. 2. Position of measurement points.

4. Results

4.1. External ambient noise

In order to evaluate the acoustical environment of the site, measurements are taken in different period on sidewalks around the building in a working day according to standard [37]. Measurements are done with B&K environmental noise platform (BZ 7210). Table 1 displays the values of the sound levels (L_{Aeq}) measured in the noisiest spots surrounding the building, as well as the range of the measured sound levels. The purpose of this measurement is to check influence of external noise (traffic noise) at the ambient noise in the building.

The values listed in the above table indicate that the building is installed in quiet zone [24].

4.2. Internal ambient noise with HVAC off

The Fig. 2 indicates the position off different selected measurement points: Point 1 is chosen in the middle of the operation room. Point 2 is facing the door to the equipment room directly overlooking the conference room. Point 5 is located approximately at the center of the conference room, while point 8 is located at the center of the podium where speakers usually are installed (see Fig. 2).

To evaluate the acoustic ambience in the selected conference room, measurements were taken of the continuous equivalent sound level. Firstly, the ambient noise was measured in the conference room when the HVAC (CTA) is OFF. The purpose of this evaluation was to check if there is any influence of internal or external noise on the acoustic ambience in the building. The Sound Pressure Level, SPL and the A and C weighted equivalent sound pressure level measured at the center of the conference room (point no. 5) are shown in Fig. 3.

Table 1
Noisiest spot level in the proximity of the building.

L_{Aeq} (dB)	L_{Amax} (dB)	L_{Amin} (dB)
57.0	74.8	49.0

These results show that ambient noise, with HVAC off, in the conference room seems to be acceptable and its value, $L_{Aeq} = 32$ dBA, is near those values recommended by national and some international standards [21–26,38–42]. This in turn encourages its use as a space dedicated to teaching. This can further be confirmed through using the NC curves method as discussed in the introduction. Fig. 4 shows that the NC which can be assigned at the measured SPL in point no. 8, with HVAC off, is NC-30. This NC value confirms the last statement for the possibilities of the use of the room as learning space.

4.3. Ambient noise in the control room with HVAC ON

The two HVAC units located in the machinery room have a self-contained refrigeration system that includes a compressor and condenser section, an evaporator coil, and a fan to circulate the air at a relatively high velocity. The combination of high velocity air movement and the noise making by different devices, such as the condenser fan, circulating air fan and compressor, results a high annoyance noise in the machinery room.

Here, the same method is adopted as for the measurement of ambient noise with HVAC off, i.e. the duration of measurements is of 3 min and the SPL with its corresponding equivalent sound level L_{Aeq}, L_{Ceq} and the range of their values.

A first measurement was taken into the control room with HVAC on, at point no. 1, and the results are shown in Fig. 5. The measurement position was chosen as equidistant from both units of the twin-HVAC system.

The measured values are displayed in Fig. 5. The values of 68 dB (A) and 78 dB(C) for the A and C weighted equivalent level respectively measured at point no. 1 is considered as relatively high. However, a simple comparison shows that the spectrum of the produced noise is similar to a standardized traffic noise given by the international standard [43].

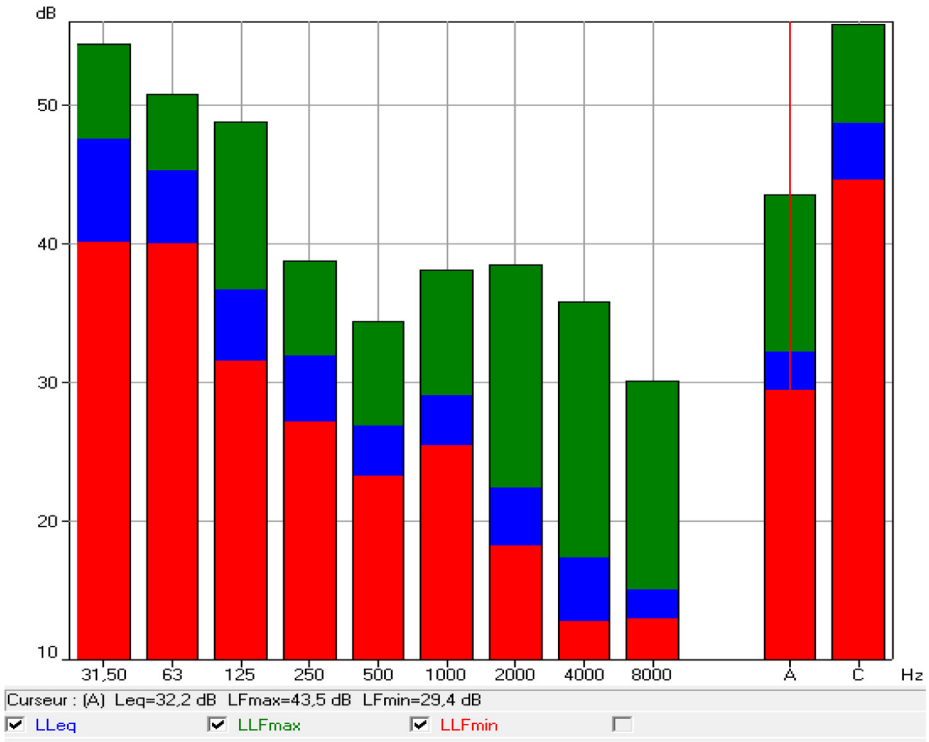

Fig. 3. Spectrum and weighted (A and C) noise level of the ambient noise with HVAC off at the measurement position, point no. 5, in the center of the conference room. The integration time was fixed at 3 min. L_{eq} is in blue, LF_{max} in green and LF_{min} in red show respectively the equivalent level, maximal and minimal measurement level in each frequency band. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

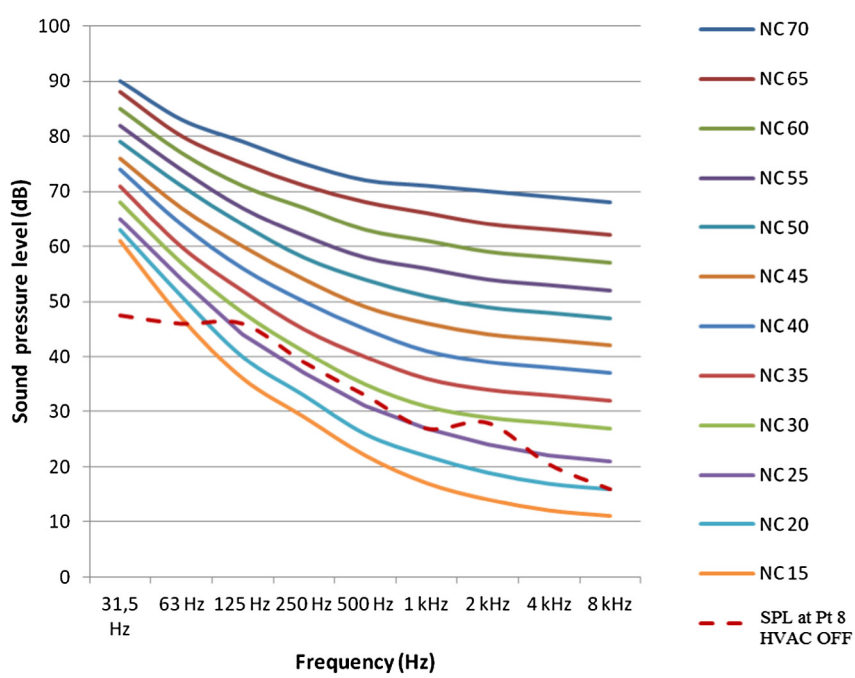

Fig. 4. SPL as measured at point no. 8 and plotted along the set of NC curves. NC values are extended to include 31.5 Hz band by comparison with balanced NC (NCB) spectrum curve. The rating of SPL is NC-30 [12].

4.4. Ambient noise in the conference room

Values of the SPL measured at different positions in the conference room with HVACs "on" are displayed in Table 2. The A- and C-weighted equivalent levels and the range of variation (min, max) are also indicated. Fig. 6 shows the variation with position of the equivalent A-filtered level LLeq.

To evaluate measurement results, we have referred to some published international standards. UK standards use the concept of suitable "indoor ambient noise levels" (IANL) for (a) clear

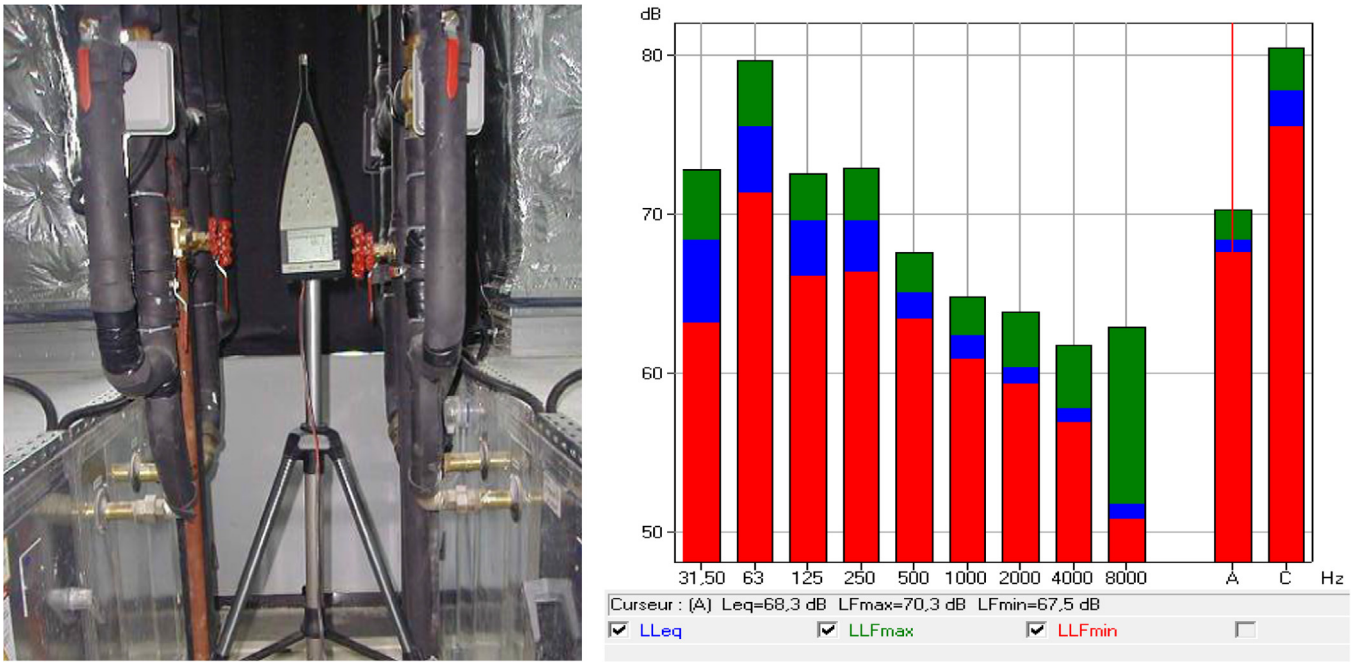

Fig. 5. Position and SPL values, even A and C weighted for ambient noise as measured with HVAC "on" at the center of control room. Integration time is fixed at 3 min. L_{eq} is in blue, L_{Fmax} in green and L_{Fmin} in red show respectively the equivalent level, show respectively the equivalent level, maximal and minimal measurement level in each frequency band. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
SPL, LAeq and it corresponding NC values calculate from [44] in different positions in the conference room.

		Frequency (Hz)									LAeq (dBA)	NC value
		31.5	63	125	250	500	1 k	2 k	4 k	8 k		
L_{eq} at measurement point no. (dB)	2	58.62	65.52	59.47	58.53	54.93	54.15	50.7	47.4	41.85	51.8	NC-53 at 1000 Hz
	3	55.52	60.89	57.89	55.98	49.97	43.87	42.94	40.63	36.87	52.5	NC-48 at 250 Hz
	4	55.25	58.52	55.84	54.12	47.81	42.98	41.76	39.74	35.77	50.9	NC-45 at 250 Hz
	5	55.87	61.36	54.67	50.76	47.62	42.81	40.97	38.64	34.04	50.0	NC-43 at 500 Hz
	6	55.74	59.28	54.47	51.16	47.44	42.62	40.36	37.82	33.24	49.6	NC-43 at 500 Hz
	7	54.62	56.18	53.24	51.65	47.41	42.11	39.7	36.77	31.7	49.1	NC-43 at 500 Hz
	8	52.47	55.18	52.21	50.21	46.79	41.55	39.2	35.7	30.04	48.4	NC-42 at 500 Hz

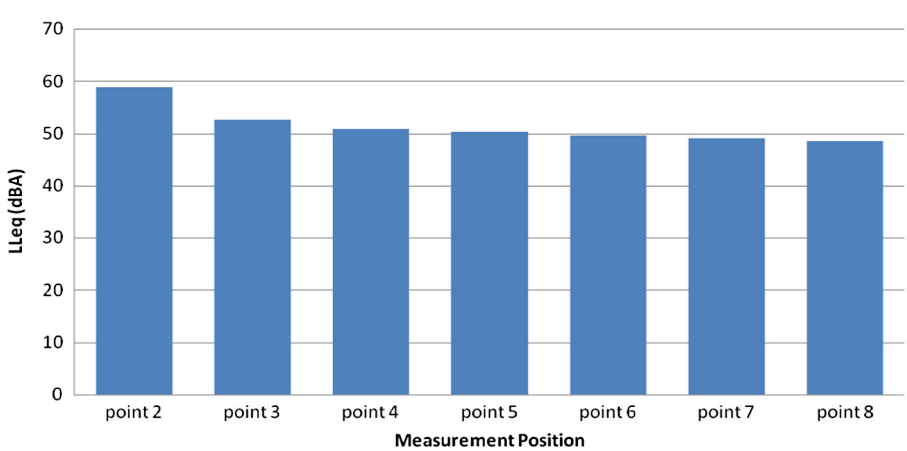

Fig. 6. Spatial variation of the LLeq (dB(A)) in the conference room.

communication of speech between teacher and student, (b) clear communication between students, and (c) learning and study activities. The IANL includes noise of building services (e.g. ventilation systems, plant, drainage, etc.) under normal use conditions. IANL values are suggested for new buildings, for refurbishments of existing buildings, and values are also given for the minimum acceptable standards for Alternative Performance Standards in new buildings [25]. Specific upper limits are given for indoor ambient noise levels expressed in terms of $L_{Aeq,30mins}$ during normal teaching hours, and these are fixed to 30 dBA and 40 dB(A) for respectively new build and for refurbished large lecture rooms (more than 50 people) [25]. The French standards [21] recommend

that the level of normalized sound pressure of the noise generated in a building by the equipment not to exceed 33 dB(A) if the equipment is operating continuously, and 38 dB(A) if it operates intermittently. In USA, the American Speech-Language-Hearing Association (ASHA) recommends that in unoccupied classrooms the noise levels must not exceed 35 dB(A) [40]. It is also mentioned that these acoustical criteria are essentially identical to the recently approved ANSI Standard on classroom acoustics ANSI/ASA S12.60 which is applicable to core learning spaces and classrooms with interior volumes not exceeding 566 m^3 [22]. This standard applies to siting and building-design-dependent sources of intrusive noise in learning spaces in schools, including noise produced by heating, ventilating, and air-conditioning (HVAC)...etc.

In the United States, it is common to use noise criterion (NC) curves to assess noise intrusion. NC curves have been established for rating indoor noise, noise from air-conditioning equipment, noise from technology used in the classroom, and other noise sources. The recommended NC level for typical classrooms is 25–30 with an equivalent sound level of 35–40 dBA [40]. Table 2 summarizes the values of SPL, L_{Aeq}, and its corresponding NC value as calculated from [44] in the conference room with HVAC "on".

4.5. Wall insulation

4.5.1. Wall elements performances

The partition wall separating technical local and conference room is consist of double well of one BA13 and a door which leads to technical room. The two partitions of the wall, BA 13 gypsum board of 12.5 mm, are mounted on the same 100 × 50 mm metal frame. The gap is filled with 10 cm of glass wool. The surface of the door is about 9% of the totality wall's surface. Dimensions of the wall are shown in Fig. 7.

For a homogeneous double BA13 wall with 10 cm of glass wool mounted on the same 100 × 50 mm metal frame, laboratory measurement yields a weighted sound reduction index R_w (C, C_{tr}) = 39 (−4, −9). The values of R, R_w C and C_{tr} are computed according to ISO 717-1 (see Fig. 8).

Concerning the door, Fig. 8 shows it acoustics data. The weighted sound reduction index is R_w (C, C_{tr}) = 23 (−1, 0) (see Fig. 9).

4.5.2. Equivalent sound transmission index

Given that the total sound power transmitted by the wall equal to the sum of the powers transmitted by each of the elements the double BA13 wall and the door (element 1 and 2 respectively), the sound transmission index of the entirely equivalent wall R_{eq} can be evaluated by introducing the notion of transmission coefficients τ of each surface element (Eq. (1)) [45]:

$$R_{eq} = 10\log\left[(S/\left(S_1 10^{-R_1/10} + S_2 10^{-R_2/10}\right)\right] \quad (1)$$

Which consequently gives an equivalent transmission index R_w (C; Ctr) = 32 (−1; −3) dB (see Fig. 10).

The sound transmission index, obtained from laboratory evaluation of product performance, take only into account direct transmissions. However, the indices used for the in situ characterization of performances reflect the totality of the transmission paths (direct and flanking).

The commonly used standards set requirements according to the type of buildings and the nature of the noise to be isolated. They are expressed in term of in situ performance characteristics. Taking into account the level of noise emitted by the equipment and its spectrum, it is clear that the elements selected for this wall do not make it able to achieve the standards requirements relating to the studied case [22,46].

4.5.3. The apparent sound reduction index of the wall

The sound insulation between rooms was measured using the actual noise (which is similar to a road traffic noise) according to ISO 16283-1. This is the preferred method when the aim of the measurement is to evaluate the performance of a whole wall including all flanking paths [47]. The measured airborne sound insulation is frequency-dependent and can be converted, according to ISO 717-1 [43] into a single number quantity: the apparent sound reduction Index (see Fig. 11).

The value obtained for the weighted apparent sound reduction index ($R'_{tr,s,w}$ = 21) dB is far below the sound reduction index when using traffic noise as a source signal ($R_{A,2} = R_w + C_{tr} = 29$ dB). Thus can give an estimation of the indirect sound transmission paths, which are contribute to worsening the situation, and which mainly achieved through flanking paths, i.e. airborne sound transmitted through supply and return air system and even structure-borne through floor, wall, and ceiling vibrations transmitted to the whole building structure. A detailed study and measurement of each transmission path is beyond the scope of this work. However, a global estimation can be given of the flanking sound reduction index paths transmissions $R_{f,w}$ using the following Eq. (2) according to the European standard EN 12354-1 [45].

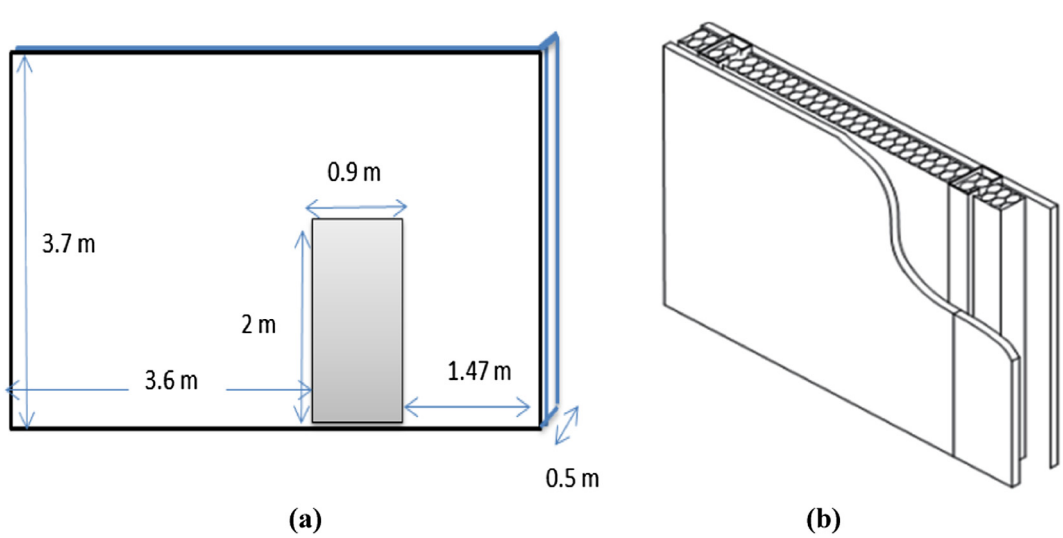

Fig. 7. (a) Dimensions of the separating wall and (b) configuration of the double well of BA13 with single light steel frame with absorbing wool.

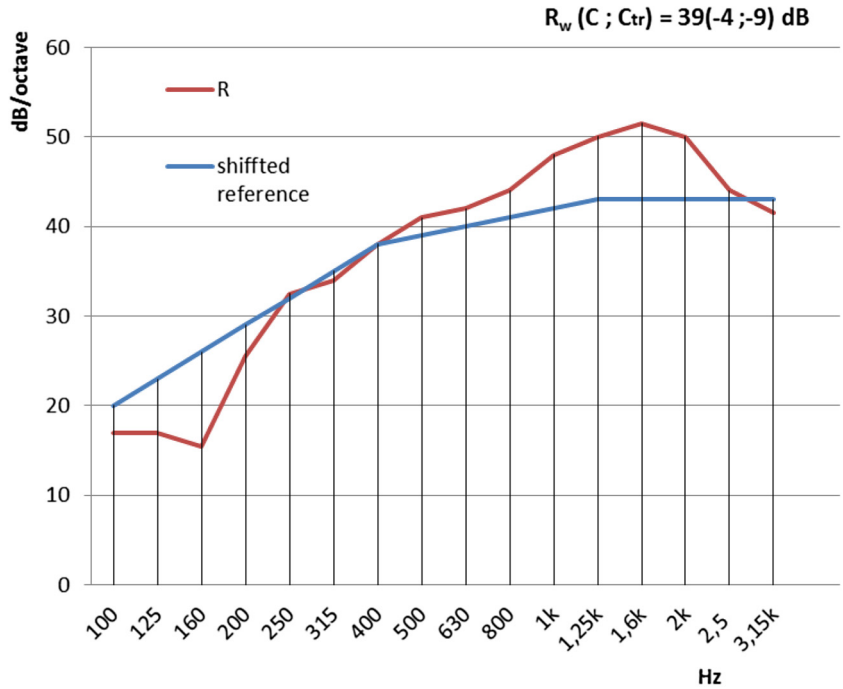

Fig. 8. Sound reduction index of the double wall partition.

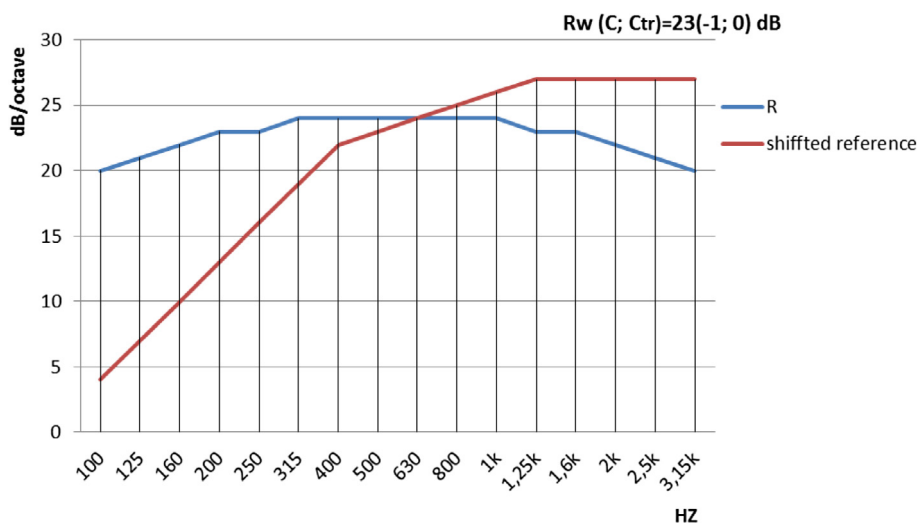

Fig. 9. Sound index transmission index of the used door.

$$R_{f,w} = -10\log\left(10^{-R'_{tr,s,w}/10} - 10^{-R_{A,2}/10}\right) \quad (2)$$

The computed value of the index $R_{f,w}$ is about 22 dB. Thus, the "in situ" performance of the separation wall is so far below the minimum recommended value which is about 50 dB [22,46].

4.6. Reverberation

The reverberation time measurements were taken at three different positions in the furnished unoccupied conference room following the method depicted in Section 2.2 and according to the standard ISO 3382-2. Three readings were taken at each point. The conference room has a volume of about 405 m^3 and can accommodate up to 70 persons, see Fig. 12. The mean values of the measured reverberation time in frequency bands are shown in Fig. 8, T_{mf} is the mid-frequency reverberation time [25] (see Fig. 13).

Several acoustical standards provide recommended values for RT depending on the use of the space. United Kingdom standard [25] sets out a minimum requirement when it comes to the acoustical design of schools. Regarding the reverberation time, this latter is quoted in terms of the mid-frequency reverberation time, T_{mf}, which is the arithmetic average of the reverberation times in the 500 Hz, 1 kHz and 2 kHz octave bands, or the arithmetic average of the reverberation times in the one third octave bands from 400 Hz to 2.5 kHz. These values are for rooms that are finished, furnished for normal use, but unoccupied. There are recommended limits of T_{mf} for new buildings and refurbishments of existing buildings, and there are also minimum acceptable standards for Alternative Performance Standards in new buildings. Although the standard does not specifically mention conference rooms, it may be concluded that the recommend value of T_{mf} for a large lecture room (seating more than 50 people) is to be equal or less than 1 s for either a new or refurbished building. The calculated T_{mf} in

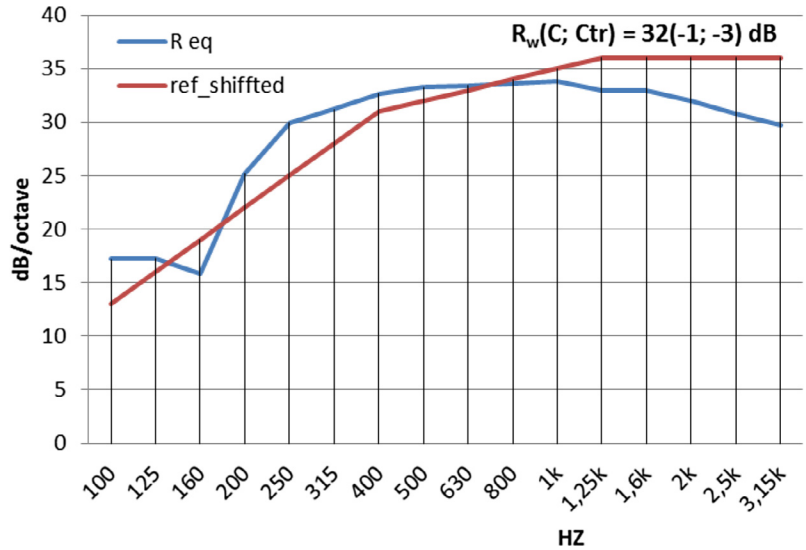

Fig. 10. The equivalent transmission index of the terraced wall.

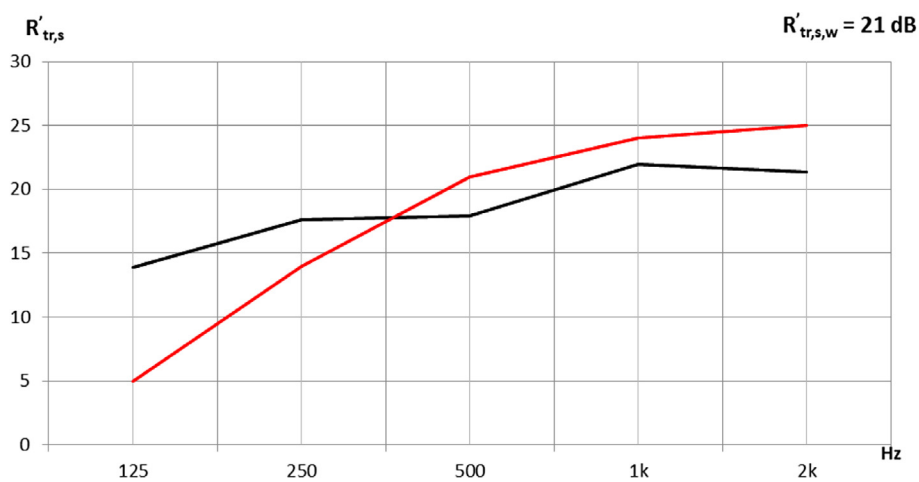

Fig. 11. Weighted apparent sound reduction index – sound insulation of façade according to ISO 717-1.

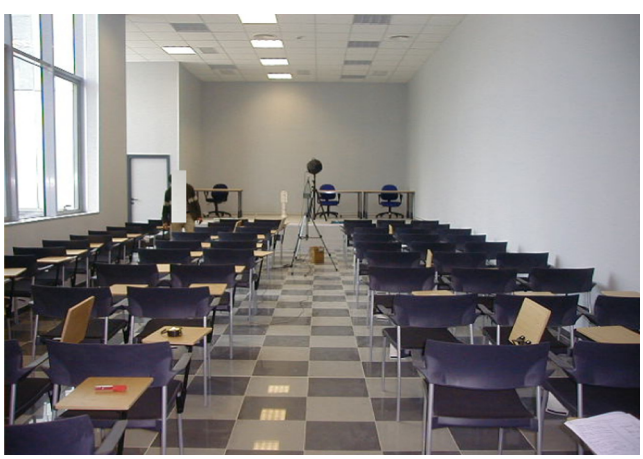

Fig. 12. View of the conference room.

the conference room is found to be about 1.27 s, which exceeds the maximum of recommend values.

ASHA [40] recommends that in the unoccupied classroom reverberation times must not surpass 0.7 s in larger rooms (283–566 m^3) or (10,000–20,000 ft^3). The French standard recommends reverberation times in the range 0.6 s < RT_{60} < 1.2 s for large rooms (volume >250 m^3). The RT being here the arithmetic mean of the RT measured in the frequency bands 500, 1000, and 2000 Hz, which is equivalent to T_{mf} [25].

The measured reverberation time shows the lack of acoustic comfort in the conference room and which is expected to affect the intelligibility of speech.

5. Discussions

5.1. Ambient noise

The ambient noise level measured in different positions of the conference room in normal use condition of HVAC system is about 49–52 dBA (NC 42–53). These levels are higher than those recommended by commonly used standards. This can be related to the high level noise emitted by the HVAC system (about 68 dBA), to the modest acoustic performance of the separate wall and to the different flanking transmissions. Moreover, the flanking transmissions and the reverberant character of both machinery and conference rooms are contribute to worsening the situation.

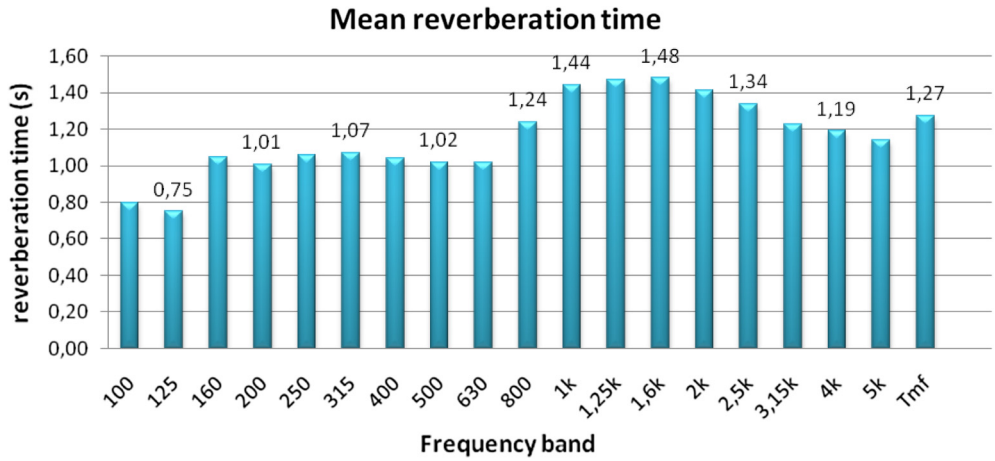

Fig. 13. Mean reverberation time by band of frequency. T_{mf} is the mid-frequency reverberation time, which is the arithmetic average of the reverberation times in the 500 Hz, 1 kHz and 2 kHz octave bands, or the arithmetic average of the reverberation times in the one third octave bands from 400 Hz to 2.5 kHz [25].

The spectrum of the produced noise, in the machinery room, is similar to their of a road traffic noise, which is characterized by a relative high level at low frequencies [43]. In addition to the guidelines given by the above cited standards, many studies have shown that low frequency noises can be disruptive and creates an uncomfortable and an inadequate environment for communication exchange. It can mask higher frequency sound, which influence on communication and on speech intelligibility especially in reverberant spaces. It can be also annoying to people who are sensitive to its effects and can produce symptoms including respiratory impairment, cardiovascular and endocrine effects and aural pain...etc. [48–50].

In engineering, many factors make low-frequencies component noise of particular concern. One of these factors it's their less attenuation by air and a low absorption coefficient at low frequencies of the commonly used porous or fibrous materials on the walls or on the floors. Similarly, walls and structures are less efficient to reduce low frequencies noises. In fact, the sound transmission index (R) decrease with decreasing frequency especially of a dip in R around a resonance and critical frequency's [51].

5.2. Technical room position and design

The position of the technical room is the main factor responsible for the acoustic discomfort felt in the conference room. This may reveal that the acoustic aspect has not been considered in the design phase. However, HVAC systems selection should be done with consideration of space planning. One of the criteria is a balance between the noise associated with the equipment in normal use conditions, and the necessary distance for reasonable noise attenuation. Thus, it's recommended to separate mechanical rooms and sensible spaces by buffer spaces such as closets, bathrooms, storage room, stairwells or elevators...etc. Also, the door to the equipment room should open to a noncritical space such a corridor or other.

When the control room cannot be established other than near a sensitive local (conference room, in our case), the terraced wall must be of sufficient performance to meet the objective set for acoustic sensitive area. If a sufficiently heavy structure cannot be used, because of the light weight, a large wide range of high performance, lightweight double, walls is available. Examples are given in (CSTB, 1999 #135), in which acoustical laboratory data of a masonry cavity wall and cavity dry partition wall on framework are given. The transmission indexes relative to a road traffic noise $R_{A,2} = R_w + C_{tr}$ are respectively about 62 and 60 dB. Knowing that the minimum "in situ" insulation or apparent transmission index, including flanking transmissions, is about 50 dB, this gives a security marge of 10–12 dB. Here, acoustic performance values are indicative and this supposes that the walls are built correctly.

Furthermore, the use of low quality separating door favors direct transmission of noise. This reduces the isolation achieved by the double walls separating the two rooms. In fact, if the location of the door can't be changed, its replacement will be inevitable. A wide variety of accredited sound control doors is available from the specialists. The choice must be made for a type whose performance is closer to or greater than that of the wall.

Other important factors can be mentioned such the low volume of the technical room and the absence of absorption surface in it. The reverberation sound energy accumulates within the local and increases the sound pressure at the inner walls of the room. Degradation of the noise reduction of the expected noise of the enclosure is implied and effect of inadequate absorption in enclosures is very noticeable. As a consequence, the wall of the technical room must be treated in a way to reduce the reverberation. One can use suspended thick surfaces of mineral wool for attenuation of the reverberation.

5.3. Reverberation in the conference room

For reverberation, it was found that the conference room offers a high reverberation time, so a poor acoustic environment. This is mainly due to the finishing materials employed, especially for the floors and facade.

A conference room should be slightly reverberant, so that the speaker does not have to raise his voice and to give some strength to speech especially the early components of its reflected part, but at the same time reverberation should not exceed the recommended values in order for the reflected sound not to overlap the direct sound and make speech unintelligible. Indeed, the room in this study is substantially a combination of three parallelepipeds and is composed of 5 pairs of parallel walls, which gives unpleasant effects due to multiple reflections. It should be noted that as the walls of the conference room are made of fine plain smooth materials, namely glass, paint plaster boards, and marble, the high acoustic reflection coefficient of these materials leads to the reverberant characteristic of the conference room. Moreover, one can notice the lack of absorbent surfaces and diffusing elements away from the speaker's location with the resulting disadvantages on

the reverberation of the room and by way reducing speech intelligibility in it, whereas their presence would offset the reflecting effects of the walls.

We note in addition the excessive length of the conference room as compared to its width. In fact, when a speaker talks, the sound waves which propagate towards the back of the room take some time before reaching it. These waves take an equally long time to reach the speaker again after reflection, and this may result in the creation of clearly audible echoes, adding more discomfort to the listener and deteriorating the quality of the message he perceives from the speaker. This effect is even more detrimental to listeners occupying the front and middle rows of the audience due to the delay gap between the direct sound and its early strong back reflection. As a consequence, the back wall of the conference room must be treated in a way to reduce the amplitude of the reflected wave or to divert it away from reaching back the front end of the room.

In fine, it necessary to precise that the execution of the refurbishment must be supervised by a specialist and must be done according to the rules of art and confirming to the manufacturer recommendations.

6. Conclusion

The acoustic measurements and physical evaluation of the lightweight structures building taken as subject of this study revealed that the acoustic aspect has not been well or fully considered in the design phase. The mains factors responsible to the acoustic discomfort are analyzed. The most important factors are the position of the machinery local and its defective design (modest acoustic performance of the terraced wall, reverberation, low volume, poor acoustic performance of the door opening directly onto the conference room ... etc.).

This indicates the need to change position of the control room or to use high acoustic insulation type wall and the door separating technical and conference rooms. In addition interior wall of room to be refurbished in order to increase their areas of sound absorption and to decrease the level of late echoes.

The architectural design discussed here is duplicated and intended to be duplicated in many other projects. Thus, the errors or acoustic deficiencies mentioned here should not be repeated in the others projects. In fact, this study has identified some design defects and proposed some rehabilitation solutions for the projects realized or in progress. This can help also to update technical specifications for future projects.

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